

ALTENRHEIN, Switzerland

WMO: 066900

Lat: 47.4836N

Lon: 9.5667E

Elev: 398

StdP: 96.63

Time Zone: 1.00 (EUC)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.4	-5.8	-12.8	1.3	-8.2	-9.0	1.8	-4.1	10.0	11.8	8.1	6.3	2.9	100	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	29.6	20.6	27.9	19.7	26.1	19.0	21.6	27.7	20.7	26.2	19.8	24.9	2.4	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	15.0	25.0	18.7	14.2	23.9	17.9	13.5	22.8	64.8	28.0	61.3	26.3	58.4	25.0	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	6.4	5.2	-10.9	32.7	2.7	2.0	-12.9	34.1	-14.5	35.3	-16.0	36.5	-18.0	37.9		
(5)			-11.4	23.7	2.9	1.2	-13.5	24.5	-15.2	25.2	-16.8	25.8	-18.9	26.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	1.9	1.7	6.3	10.4	13.9	18.4	19.9	19.6	15.4	11.0	6.1	2.7
	DBStd	7.39	3.90	4.29	3.20	3.69	3.50	3.18	2.89	2.91	2.83	3.22	3.38	3.54
	HDD10.0	1028	252	232	119	39	7	0	0	0	0	28	125	227
	HDD18.3	2979	509	465	373	241	142	39	17	20	94	228	367	484
	CDD10.0	1271	1	0	4	50	128	251	308	298	164	58	8	1
	CDD18.3	179	0	0	0	1	5	39	66	60	8	0	0	0
	CDH23.3	1220	0	0	0	13	46	295	464	369	31	1	0	0
	CDH26.7	287	0	0	0	2	4	76	121	85	1	0	0	0
Wind	WSAvg	2.0	2.0	2.1	2.2	2.2	2.1	2.1	2.0	1.8	1.7	1.7	1.8	2.0
Precipitation	PrecAvg	1400	69	77	88	101	138	171	168	170	124	96	104	91
	PrecMax	1799	150	210	177	188	321	314	311	291	200	208	238	192
	PrecMin	1067	7	21	30	32	19	65	65	49	57	9	7	10
	PrecStd	155	40	41	37	42	67	61	65	60	39	43	50	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	14.9	18.3	25.0	27.1	30.9	32.3	31.1	26.0	22.2	19.7	17.1
		MCWB	9.6	7.8	10.3	13.9	17.2	21.1	21.5	21.2	19.2	13.1	10.7	8.2
	2%	DB	10.2	11.3	16.1	21.9	24.3	28.8	29.7	29.0	24.1	19.0	15.9	11.9
		MCWB	7.1	7.3	9.1	12.7	15.8	19.9	20.3	20.8	18.3	14.8	11.4	8.3
	5%	DB	8.2	9.0	14.2	20.0	22.6	26.8	27.8	27.1	22.2	17.1	12.7	9.4
		MCWB	6.1	5.9	8.7	12.1	15.1	19.3	19.7	20.2	17.5	14.2	10.0	6.9
	10%	DB	6.8	7.0	12.1	17.4	20.8	24.5	25.9	25.2	20.5	15.5	10.8	7.8
		MCWB	5.1	4.8	7.9	11.2	14.4	18.2	18.8	19.4	16.7	13.2	8.8	5.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	9.7	11.4	15.1	18.7	23.2	22.7	22.6	20.0	16.1	12.6	10.1
		MCDB	14.0	13.7	16.9	22.5	24.7	29.8	29.6	28.5	24.3	19.0	15.8	14.6
	2%	WB	7.8	7.5	10.2	13.7	17.2	21.1	21.3	21.6	18.9	15.1	11.4	8.7
		MCDB	9.7	10.7	15.0	20.0	22.5	27.3	27.6	27.4	23.0	17.7	14.7	11.5
	5%	WB	6.3	6.1	9.1	12.8	16.1	19.8	20.3	20.7	18.0	14.3	10.2	7.3
		MCDB	7.8	8.5	13.0	18.6	20.8	25.5	26.1	25.9	21.9	16.5	13.0	9.3
	10%	WB	5.2	4.8	8.2	11.8	14.9	18.8	19.4	19.9	16.9	13.5	9.1	6.1
		MCDB	6.5	6.8	11.8	16.7	19.3	23.7	24.8	24.5	20.0	15.4	10.8	7.4
Mean Daily Temperature Range	5% DB	MDBR	5.2	6.2	8.9	9.7	9.2	9.0	9.2	9.2	8.2	6.9	5.4	5.4
		MCDBR	7.7	10.4	13.5	14.2	12.8	12.9	13.0	12.2	10.4	9.4	9.5	8.4
		MCWBR	5.5	6.4	7.8	7.1	6.0	6.1	5.7	5.7	5.6	5.7	5.6	5.5
	5% WB	MCDBR	7.4	10.0	11.5	13.1	11.4	11.7	11.8	11.4	10.0	8.5	8.7	8.0
		MCWBR	5.5	6.3	6.8	7.1	5.7	5.9	5.4	5.6	5.6	5.6	5.5	5.5
Clear-Sky Solar Irradiance	taub	0.291	0.317	0.358	0.399	0.414	0.420	0.422	0.413	0.383	0.351	0.318	0.287	
	taud	2.410	2.336	2.307	2.241	2.251	2.276	2.268	2.308	2.371	2.446	2.476	2.429	
	Ebn at Noon	795	842	858	852	851	846	838	829	822	796	757	755	
	Edh at Noon	68	90	109	128	132	130	129	119	102	81	64	59	
All-Sky Solar Radiation	RadAvg	1.15	2.00	3.19	4.32	4.98	5.70	5.52	4.82	3.61	2.28	1.26	0.95	
	RadStd	0.13	0.27	0.42	0.61	0.51	0.54	0.51	0.50	0.38	0.25	0.15	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (111 neighbors)		N/A	N/A	N/A	N/A	+0.44	+0.37	N/A	N/A	+53	+14

Nomenclature: See separate page